

AMENDMENT NO. 2 MAY 2013
TO
IS 10491 : 1983 SPECIFICATION FOR MUZZLE LOADING SHOT GUNS,
SINGLE AND DOUBLE BARREL

(Page 2, clause 3) — Substitute the following for the existing clause:

‘3 MATERIAL

SI No.	Name of Parts	Material
i)	Barrel	IS 5517 : 1993 ‘Steel for hardening and tempering’ grade 42Cr4Mo3or 40C8 or 45C8 or 50C8. If made by forging then IS 1875 : 1992 ‘Carbon steel billets, blooms, slabs and bars for forging’, Grade 45C8 or 55C8. Heat treatment -To be heat treated to achieve mechanical properties given below: UTS=700 to 850 MPa 0.2 Percent proof stress=480 MPa, <i>Min</i> IS:13015, C1S1R1 Inclusion rating- As per IS 4163 : 2004 Thick series : Nil Thin series : 1.0 (<i>Max</i>) Hardness – 210 to 270 HV
ii)	Body	IS 5517 : 1993 Grade 55C8 or 40Ni 14 or IS 3930 : 1994 Flame and induction hardening steels Grade 37 Mn 6 and Si shall be 0.10 to 0.35 percent.
iii)	Trigger	IS 5517 : 1978 Grade 35Mn6Mo4 or 42Cr4Mo2 or 40Cr4 or IS 4368 : 1967 ‘Alloy steel billets, blooms and slabs for forgings for general engineering purposes’ Grade 40Cr I or 40CrI Mo28. Heat treatment -To be heat treated to achieve mechanical properties given below: UTS = 800 MPa <i>Min</i> 0.2 Percent proof stress=600 MPa <i>Min</i> Elongation on $5.65\sqrt{A}$ = 16 percent <i>Min</i> Izod Impact=55J Hardness=250 HV generally and 400 HV locally at the catch point
iv)	Sear	IS 5517 : 1993 Grade 31NiOCr3Mo6 or IS 3431 : 1982 ‘Steel for volute, helical and laminated springs for automotive suspension’ Grade 50Cr4V2 or 55S17 or 60Si7 or 65Si7 or IS 3885 (Part I) : 1992 ‘Steel for manufacture of laminated springs (railway rolling stock): Part 1 Flat sections (<i>first revision</i>)’ Grade 55S17. Heat treatment - To be heat treated to achieve hardness of 350-450 HV generally and 600-700 HV locally at catch point.
v)	Hammer / Pin Firing	IS 5517 : 1993 Grade 55C8 or 40Cr4 or 37C 15. If made by forging of IS 1875 : 1992 Grade 55C8 or IS 4368:1967 Grade 40Cr 1, IS 2004 : 1992 Grade 55C8 Heat treatment - To be heat treated to achieve hardness of 350-450HV generally and 500-550 HV locally at striking face.
vi)	Main spring	IS 3431 : 1982 ‘Steel for manufacture of volute, helical and laminated springs for automotive suspension (<i>second revision</i>)’ Grade 55S17 or 60S17 or 65Si7or IS 3885 (Part I) : 1982 ‘Steel for the manufacture of laminated springs (railway rolling stock): Part 1 Flat sections (<i>first revision</i>)’ Grade 55S17 Heat treatment - To be heat treated to achieve hardness — 450-550 HV.

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vii)	Fastener fore-end	IS: 7887-1992, Grade-8 with hardness restricted to 115 to 160 HV
viii)	Spring lever top	IS 3431 : 1982 Grade 55Si7 or 60S17 or 65S17 or IS 3885(Part 1) : 1992 Grade 55S17. Heat treatment - To be heat treated to achieve hardness of 500 -550 HV
ix)	Catch hook	IS 5517 : 1993 Grade 55C8. Heat treatment - To be heat treated to achieve hardness of 400-450 HV.
x)	Stock butt and stock fore-end	IS 7549 : 1975 'Timber half wrought for sporting rifles'

(Page 3, clause 4.2) — Insert the following after the existing clause:

- 4.2** a) Ensure the shot gun is unloaded and it is safe. Following action needs to be taken:
- 1) Safety lever should be placed in safety mode/half cock position.
 - 2) Look through the barrel/barrels and ensure that the chamber is empty.
- b) Ensure that the acceptance proof marks are stamped/engraved on the location as indicated below:
- Barrel/Barrels
Proof mark
SBML — Ashoka Lions
DBML — Letter 'P' surmounted by an Ashoka Chakra & Ashoka Lions in both barrels.
- c) Registered number of shot guns should tally in each of its components where the registered number exists. Normally in the shot gun the registered number exists in the following components:
- 1) Body
 - 2) Barrel/Barrels
- d) Before firing maintenance (before firing a shot gun the following action should be taken):
- 1) Ensure that the gun is not loaded.
 - 2) With a piece of cloth wipe out oil from external parts.
 - 3) Using flannelette clear the bore, dry and inspect the bore.
 - 4) Check functioning of trigger and firing pin. Firing pin should not be sticking jammed
 - 5) Check the functioning of safety lever.
- e) After firing Maintenance:
- 1) Barrel:
 - i) Clean externally with a dry cloth and remove oil.
 - ii) Run hot water through the barrel for twenty minutes.
 - iii) With the help of a cleaning rod and flannelette clean the bore and dry.
 - iv) When perfectly dry, apply lightly gun oil internally using the rods and flannelette.
 - v) Apply oil lightly on external surface.
 - vi) Hammer and pin firing to be cleaned checked and oiled.
 - vii) Check functioning of trigger, safety lever and firing pin.

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2) Butt:

- i) Clean with dry cloth.
- ii) Half yearly or any other occasion when it is essential to remove furniture components apply linseed oil with a piece of cloth and allow it to dry for 03h. Care must be taken not to allow the linseed oil to touch the metal parts. After application of linseed oil the weapon must not be polished.'

(Page 4, clause 8, lines 2 and 4) — Substitute 'IS 6005 : 1998' for 'IS 6005 : 1970'.